DATABASE AUTOMATION FOR MANUAL LOGBOOK RECORDS REPORT

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Freelance

Data: Manual Logbook

Tools: Excel

EXECUTIVE SUMMARY

176	331	80
Total Unique Students	Total Students Record	Total Number of Customers

This report covers a project where I converted Crown Hope's manual logbook records into proper Excel databases. The business runs computer and engineering training alongside sales of laptops and accessories like chargers and mouse, but had no digital system in place before this. Everything, from student fees to customer payments, was written by hand in one logbook with nothing separating the two.

Since the logbook didn't clearly mark which entries were students and which were customers, I worked out the difference by looking at the amount paid, since student fees and customer purchases fell into noticeably different ranges. From there I built separate relational databases: one for students, one for customers, one for inventory, and a summary sheet pulling everything together. Alongside the logbook, students had also filled out registration forms containing their personal details, so I developed a macro to automate the entry of this information into the student database, reducing reliance on paper records and cutting down on manual retyping.

The data covers **October 2016 through December 2025**. Once organised, the student database showed **331 payment entries**, but only **176** of those were actually **unique students**, since many paid in installments and appeared more than once under the same name. On the customer side, there were **80 unique customers** recorded for laptop and accessory purchases.

The client now has a proper record system instead of a paper logbook, making it far easier going forward to track who has paid, who's still owing on an installment plan, what's been sold, and what's left in stock.

PROBLEM STATEMENT

The business relied entirely on a manual paper logbook to record all payments, with no system in place to distinguish between student and customer transactions or to track partial payments made in installments. As a result, it was difficult to determine which students had fully paid, which still owed a balance, and how much had actually been collected over time. Names appearing multiple times in the logbook due to installment payments made it nearly impossible to identify actual unique students versus repeat entries, leading to unreliable records and no clear way to monitor outstanding payments or verify total enrollment numbers.

DATA OVERVIEW

This project focuses on transforming manual logbook records into a structured Database Management System (DBMS) for a small computer sales and training business, to ensure data safety, accuracy, and long-term usability for analysis and decision-making.

The primary goal was to convert messy, unstructured manual records into well-organized relational databases using Microsoft Excel. A macro was developed to automate the entry of student information from filled registration forms directly into the database, reducing reliance on paper records and improving efficiency and reliability.

Data Quality Issues & Decisions

Issue Found	Decision Made
No clear separation between student and customer entries in the logbook	Categorised entries by amount paid, since student fees and customer payments fell into distinct ranges
Same student name appearing multiple times	Attributed to installment payments; used this to distinguish unique students (176) from total payment entries (331)
Student personal details existed only on paper forms	Developed a macro to transfer form data into the student database instead of manual retyping

METHODOLOGY

A five-stage pipeline using Excel was adopted to ensures reproducibility and separation of raw from analytical data.

Stage	Key Decision
Data Collection	Gathered raw data from two sources, the manual logbook (October 2016 to December 2025) and paper registration forms filled by students
Data Classification	Since the logbook mixed student and customer entries with no labels, records were classified based on amount paid, as student and customer payments fell into distinct ranges
Database Design	Built relational databases in Excel, linked through shared identifiers, to keep raw data separate from analytical output. Six databases created: Student Data, Student Payment, Customer Data, Customer Payment, Laptop & Accessory Inventory, and Summary.
Automation	Developed a macro to transfer student information from registration forms directly into the student

	database, reducing manual retyping and transcription errors
Validation and Analysis	Used SUMIFS to total previous installment payments per student and determine actual amount paid versus outstanding balance owed. Applied IF formulas to flag payment status (e.g., paid in full vs owing). Used XLOOKUP to link student names and customer names to the Summary database, making it easy to track each individual's records across the system

KEY FINDINGS

1. Student Information Records
Manual student forms contained incomplete and inaccurate information. Some students refused to provide key details such as date of admission and phone number, which are essential for tracking training duration, completion dates, and overall student progress. Others filled the forms incorrectly, mostly due to a lack of understanding of what the form actually required.

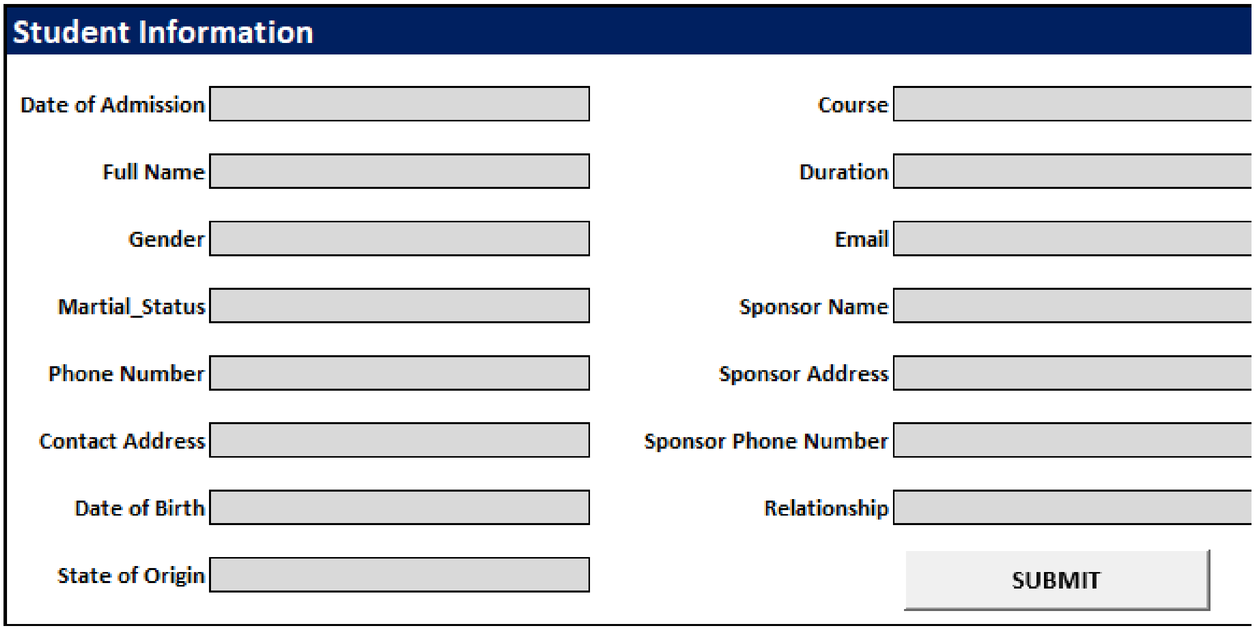
2. Student Payment Records
Payments were recorded manually, making it difficult to track which students had fully paid and which were still owing, risking missed follow-ups and potential loss of outstanding fees.

3. Sales Records
Sales data was recorded in logbooks, making it difficult to track customer patronage and churn, and to identify customers with outstanding balances, limiting the business's ability to spot repeat customers or catch lost business early.

4. Certificate Management
There was no certification record, making it impossible to confirm which students had collected their certificates or whether a certificate was being reissued. This could result in unplanned costs for the business.

5. Inventory Management
There was no proper tracking of laptops and accessories sold versus items available in stock. This could lead to incorrect sales decisions and financial losses.

Visual Insight & Dashboard



RECOMMENDATION

1. Provide proper orientation to students on how to fill forms and why accurate information matters.
2. Adopt online forms (e.g., Google Forms) with mandatory fields to improve data accuracy and completeness going forward.
3. Upgrade payment tracking from formula-based (SUMIFS/XLOOKUP/IF) to a fully automated VBA-based payment system for real-time balance updates.
4. Track payment status clearly to identify customers with outstanding balances.
5. Assign an administrator responsible for consistently updating records to maintain professionalism and accuracy.
6. Continue linking inventory data with sales records to prevent stock errors and reduce

CONCLUSION

This project was carried out to convert Crown Hope's manual logbook records into a structured Database Management System, improving how the business tracks students, customers, payments, certificates, and inventory. Relational databases were successfully created for Student, Customer, Inventory, Summary, and Certificate records, and all existing manual logbook entries were accurately migrated into the new system.

During data entry, several gaps and inconsistencies in the previous manual process were identified, including incomplete student information, difficulty tracking installment payments, no visibility into customer patronage or churn, and no formal way to track certificate issuance or inventory. A VBA macro was developed to automate the transfer of student registration details into the database, reducing manual retyping, while formulas such as SUMIFS, XLOOKUP, and IF were used to track payment status and link records across the system.